

DISCUSSION NOTE: TWO-PHASE DEAD CORE FOR THE FRACTIONAL p -LAPLACIAN STATUS, ONE QUESTION ABOUT *IMPROVED REGULARITY FOR A NONLOCAL DEAD-CORE PROBLEM*, AND A PROPOSED ARGUMENT

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1. STATUS IN ONE PARAGRAPH

For $(-\Delta_p)^s u = u_+^\gamma - u_-^\gamma$ in B_1 ($0 < \gamma < p-1$, $p \geq 2$), the target is growth of order $\alpha = \frac{sp}{p-1-\gamma}$ at points with $u = |Du| = 0$. The scaling exponent and the vanishing of the right-hand side along blow-ups (via $(p-1)$ -homogeneity, replacing linearization) are done. Following Correa–dos Prazeres §6, the Liouville theorem is reformulated with a vanishing 1-jet hypothesis, avoiding affine subtraction; the jet-to-growth lemma transplants verbatim. The interior C^{1,α_0} input is Giovagnoli–Jesus–Silvestre (arXiv:2509.26565); their own remark suggests the needed $f \in L^\infty$ extension is feasible for $p < \frac{1}{1-s}$ (forcing $s > \frac{1}{2}$, as in your paper). Two caveats found: their constants blow up as $s \nearrow 1$, so the local two-phase theorem should be proved directly rather than by the $s \nearrow 1$ limit; and the tail bookkeeping below. Since we last spoke I have checked the proposed (4.12) repair step by step (§3, now sound in outline) and intersected the constraint regions (§4): the explicit admissible region is nonempty, and the one genuinely binding constraint turns out to be the size of the C^{1,α_0} exponent.

2. THE QUESTION ABOUT (4.12)

In the proof of Theorem 4.1, the growth bound (4.11) for $w_k = v_k(\cdot + h) - v_k(\cdot)$ rests on (4.8), which is justified for $|x| \leq \frac{1}{2r_k}$ (as (4.7) requires $r_k R < 1/2$). Beyond that radius, u_k is only bounded, so $|w_k| \leq 2(\theta_k(r_k)r_k^{\frac{2s}{1-\gamma}})^{-1}$. Passing (4.12) to the limit via [CS, ARMA 2011] requires $w_k \rightarrow w_0$ in $L^1(\mathbb{R}^n, \omega)$ *globally* (their Remark 4.4 shows local convergence cannot suffice), and the outer region contributes $\sim \theta_k^{-1} r_k^{-\frac{2s\gamma}{1-\gamma}}$, which is bounded only under a lower bound on θ_k that (4.5) does not seem to provide. **Question: how is the far field handled in (4.12)?** I may well be missing something standard.

3. A PROPOSED ARGUMENT (CHECKED STEP BY STEP; SOUND IN OUTLINE)

Truncate: $\tilde{w}_k = w_k \chi_{B_{\rho_k}}$, $\rho_k = \frac{1}{4r_k}$. Then $(-\Delta)^s \tilde{w}_k = f_k + H_k$ in B_R , where H_k is the far field as a datum. The point is that only the *oscillation* of H_k matters:

- (1) $\|\nabla H_k\|_{L^\infty(B_R)} \lesssim C_h \rho_k^{\alpha-2s-2} + \theta_k^{-1} r_k^{2s+1-\alpha} \rightarrow 0$, the second exponent being positive iff $\gamma < \frac{1}{1+2s}$ (the standing assumption). Hence $H_k = a_k + o(1)$, a_k constant.
- (2) a_k is bounded *a posteriori*: $\tilde{w}_k$ has uniform local regularity and uniform L_s^1 norm, so $(-\Delta)^s \tilde{w}_k$ is uniformly bounded at a fixed point, while $f_k \rightarrow 0$.
- (3) Stability now applies and gives $(-\Delta)^s (v_0(\cdot + h) - v_0) = a(h)$ with $a(h)$ finite.
- (4) The cocycle identity $w^{h_1+h_2} = w^{h_1}(\cdot + h_2) + w^{h_2}$ forces $a(h) = b \cdot h$.

- (5) Second differences: $(-\Delta)^s \Delta_h^2 v_0 = 0$ entire, with growth $(1 + |x|)^{\alpha-2}$; Theorem 3.1 ($\nu = 1$) makes $\Delta_h^2 v_0$ affine for every h , so v_0 is a polynomial of degree ≤ 2 (by the growth $\alpha < 1 + 2s$), and the vanishing 2-jet at the branching point gives $v_0 \equiv 0$.

No lower bound on θ_k is needed. The step I worried about most is (2): it is *not* circular — the bound on $(-\Delta)^s \tilde{w}_k$ at a fixed point splits into a local singular part (controlled by the uniform interior $C^{2s+\gamma}$ estimate) and a tail (controlled using only the growth bound $|w_k(y)| \leq C_h |y|^{\alpha-1}$ on the truncated support, finite iff $\gamma < \frac{1}{1+2s}$), neither of which uses a_k or any bound on θ_k ; a_k is then solved for. The truncation is precisely what manufactures the global L_s^1 convergence that (4.12) needs.

The same far-field-as-datum mechanism carries to $p \neq 2$ for steps (1)–(3): the increments solve the *linear* equation $L_{K_{v_k}} w = g$ with the kernel $K_v \simeq |v(x) - v(y)|^{p-2} |x - y|^{-n-sp}$ of GJS §7, and the oscillation computation produces the additional condition $\gamma < \frac{p-1}{1+sp}$ (which coincides with $\frac{1}{1+2s}$ at $p = 2$, stricter for $p > 2$). Steps (4)–(5), however, use translation invariance of $(-\Delta)^s$, which the v -dependent kernel K_v does *not* have; so for $p \neq 2$ I would drop the constant- $a(h)$ route and apply the vanishing-jet Liouville to v_0 directly (where $a(h)$ never appears) — this is one of the points I would like to confirm with you.

4. WHAT I WOULD LIKE TO DISCUSS

- (1) Whether the (4.12) argument of §2–§3 is right, or whether the far field has a standard justification I missed.
- (2) Whether, for $p \neq 2$, the vanishing-jet Liouville on v_0 indeed removes the need for steps (4)–(5) (which rely on the translation invariance lost when $p \neq 2$).
- (3) The one genuinely binding constraint: after intersecting the tail, jet-order and datum conditions (all explicit and jointly nonempty; $\nu = 1$ alone suffices for $p > 2$, the window staying open up to $s = 1$), the binding requirement is $\alpha < 1 + \alpha_0$ with α_0 the GJS exponent. Do we state the theorem with α_0 as an abstract constant, or try to extract an explicit lower bound $\alpha_0(n, s, p)$ from their proof? Also: the kernel K_v degenerates where v is flat (worst at the branching point) — is there a clean way to see v_0 is nondegenerate on an open set?
- (4) Proving the local two-phase p -dead-core theorem directly (the $s \nearrow 1$ route is blocked by the non-uniformity of the GJS constants); it does not appear in the literature and may stand alone.
- (5) Whether to include the $f \in L^\infty$ extension of GJS for $p < \frac{1}{1-s}$ as a section of our paper.

All details, computations and sources are in the project repository: github.com/hikmatulloismatov1999/fracti